UNIVERSITY OF WEST GEORGIA

Main Campus · Dr. James 'Earl' Perry College of Mathematics, Computing, and Sciences · Computing, Analytics, and Modeling

Intelligent Systems

CS-3270

Spring 2026 Section 01 3 Credits 01/10/2026 to 05/12/2026 Modified 01/03/2026

Description

Application and survey of problem-solving methods in artificial intelligence with emphasis on heuristic programming, production systems, neural networks, agents, social implications of computing, and professional ethics and responsibilities.

Requisites

Prerequisites:
CS 1302 Minimum Grade: B and (MATH 1634 or MATH 1501)
Corequisites:

Contact Information

Instructor Name: Dr. Md Shirajum Munir

- **Email**: mmunir@westga.edu
- **Office Phone:** (678) 839 6664
- **Office:** Technology Learning Center (TLC) 2248, 210 W Georgia Dr, Carrollton, GA 30117

Office Hours (Spring 2026):

Monday and Wednesday (In-person): 1:00 PM to 2:00 PM (Eastern Time)

Tuesday and Thursday (Online): 12:00 PM to 1:00 PM (Eastern Time)

Others: By appointment

Online meeting schedule instruction: At least email by Monday 12:00 PM for Tuesday schedule and email by 12:00 PM Wednesday for Thursday schedule. Students can schedule online meetings during in-person/online office hours based on prior discussions or email conversations. The team meeting link will be sent to join.

Note: If any of the above time slots do not work, then drop an email to schedule an online/in-person meeting. The schedule will be adjusted based on availability.

The official communication (located on page 41) method is through your university email account. Emails sent through the CourseDen class website may not receive a response since this course usages Moodle, the Computer Science Content Management System, which can be found here: CS3270: Intelligent Systems (Section 1) [S26] (https://courses.cs.westga.edu/course/view.php?id=1234).

Email Format:

- Please always use the UWG email.
- Subject [Course Code] [Your Name] Your Subject
  - Example: [CS3270] [Name] Needs to discuss on assignment 1
- Signature: Your Full Name and UWG ID Number
  - Example:
    - First Name Last Name
    - 917-xxxxxx

Meeting Times

Class Time: Monday and Wednesday 10:30 AM to 11:45 AM (From 01/10/2026 to 05/12/2026)

Location (Room): Technology Learning Center (TLC) 1210

All classes and lectures will be held in person. If an exception occurs, students will be notified through a Moodle announcement. For instance, in cases such as conference travel, severe weather, or other unavoidable circumstances, the instructor will provide online materials or additional study instructions as appropriate.

Attendance Policy:

All students are expected to attend and actively participate in every class and lecture. The instructor will closely monitor in-class engagement (e.g., discussions, conversations, Q&A sessions) and attendance. If a student cannot attend a class due to unavoidable circumstances, they must provide proper justification, such as a medical certificate or other valid documentation.

Missing in-class activities (e.g., sudden/surprise quizzes or coding exercises based on previous lectures) will affect the final grade. Throughout the semester, there will be five such sudden/unannounced class activities, each worth 1% of the final grade. Missing any of these will result in a 1% deduction from the total score for each missed activity (please see the Evaluation section for more details on class activities).

Materials

There is no mandatory required textbook for this class. Reading assignments will come from freely available online resources. References (links, documents, videos) will be provided as the course unfolds.

All content for this course is located in and can be accessed via Moodle, the Computer Science Content Management System, which can be found here: CS3270: Intelligent Systems [S25] (https://courses.cs.westga.edu/course/view.php?id=1234).

Books and Resources that students can use for their own interest (not mandatory):

Analytics, Data Science, & Artificial Intelligence: Systems for Decision Support

Author: Ramesh Sharda, Dursun Delen, Efraim Turban
Publisher: Pearson
Edition: 11th
Year: 2019
ISBN: 978-0135192016
Optional

Python Machine Learning

Author: Raschka Sebastian, and Vahid Mirjalili
Publisher: Packt
Edition: Third
Year: 2019
ISBN: 978-1789955750
Optional

https://github.com/rasbt/python-machine-learning-book-3rd-edition

Artificial Intelligence: A Modern Approach

Author: Stuart Russell and Peter Norvig
Publisher: Pearson
Edition: Third
Year: 2009
ISBN: 978-0136042594
Optional

Outcomes

The students will understand the basic components, principles, and process of designing Intelligent Systems. In particular, the students will be able to identify the problems of Intelligent Systems and their solutions. At the end of this class, students should be able to:

- Understand the Intelligent Systems and their implementation process.
- Understand the role of artificial intelligence (AI), machine learning (ML), data science, heuristic programming, and agent-based programming for Intelligent Systems.
- Apply heuristic and optimal search methods to solve a problem.
- Use ML methods to solve a problem by following best practices.
- Experiential learning of Intelligent Systems based on AI/ML, sensors, Raspberry PI, and corresponding software.

- Recognize professional and ethical responsibilities in relation to the design, implementation, and use of intelligent systems.
- Communicate technical results to non-technical audiences clearly and correctly, in writing and verbally.
- Effectively function as a member of a team to apply fundamental concepts of AI/ML to solve real-world intelligent systems problems.

Evaluation

Student Tasks and Evaluation Weight:

Task	Weight
Syllabus Quiz	1%
Assignments (5x6) [each of them will be 6% points] [coding, report writing, and demonstration in the classroom]	30%
Intelligent Systems Topic Presentation	5%
Class Activities (5x1) [each of them will be 1% points] (sudden class quizzes or coding practices based on previous class lecture)	5%
Final Group Project Proposal Presentation	5%
Exam (3x13) [each of them will be 13% points]	39%
Final Group Project Demo Presentation and Submission	15%

Note: Detailed weight distribution will be described for each task during the individual task announcement in Moodle.

Criteria

Understanding topics, assignment completion, exam/quiz, project demonstration, in-class activities, and feedback. The course will be evaluated through the assignments described in the Assignments section.

The grade will be determined based on the total points earned at the end of the semester. The final grade will be calculated as follows:

90% - 100%	A
80% - 89.99%	B
70% - 79.99%	C
60% - 69.99%	D
0% - 59.99%	F

Assignments

Task	Submission Open	Submission Deadline
Syllabus Quiz	January 12, 2026, Monday 10:00 AM	January 18, 2026, Sunday 11:55 PM
Assignment 1	January 19, 2026, Monday 10:00 AM	January 25, 2026, Sunday 11:55 PM
Assignment 2	February 9, 2026, Monday 10:00 AM	February 15, 2026, Sunday 11:55 PM
Intelligent Systems Topic Presentation	NA	February 18, 2026, Wednesday 10:30 AM (In person)
Exam 1 [Exam format will be written and/or MCQ. Detailed instructions will be provided in Moodle based on course progress.]	NA	February 25, 2026, Wednesday 10:30 AM (In person and closed book)
Assignment 3	March 2, 2026, Monday 10:00 AM	March 08, 2026, Sunday 11:55 PM
Final Group Project Proposal Presentation [Instruction will be provided based on course progress]	NA	March 25, 2026, Wednesday 10:30 AM (In person)
Assignment 4	March 30, 2026, Monday 10:00 AM	April 05, 2026, Sunday 11:55 PM

Exam 2 [Exam format will be written and/or MCQ. Detailed instructions will be provided in Moodle based on course progress.]	NA	April 08, 2026, Wednesday 10:30 AM (In person and closed book)
Assignment 5	April 20, 2026, Monday 10:00 AM	April 26, 2026, Sunday 11:55 PM
Final Group Project Demo and Submission	April 06, 2026, Monday 10:00 AM	Presentation: April 29, 2026, Wednesday 10:30 AM (In person) Submission: April 29, 2026, Wednesday 11:55 PM
Exam 3 [Exam format will be written and/or MCQ. Detailed instructions will be provided in Moodle based on course progress.]	NA	Wednesday, May 6, 12:00-2:00 PM (In person and closed book)

All schedules follow Eastern Time.

This is an initial schedule and plan for assignments and exams. The schedule is subject to change based on class progress and other circumstances. Any changes will be notified via Moodle (https://courses.cs.westga.edu/course/view.php?id=1234) announcements.

Note: Detailed weight distribution will be described for each task during the individual task announcement in Moodle. Group creation and final project description will be announced during the course progress.

Schedule

The initial schedule of this course is given below, which may change based on students' progress and their feedback during the course. For **important dates (drop-add [Jan 10 - 16], withdrawal last day [Mar 25])**, visit the **Spring 2026 calendar** (https://www.westga.edu/student-services/registrar/term-spring_2026_sessions_calendar.php).

Weeks	Activities
Week 1: Jan 10 - Jan 16	• Syllabus Review and Course Overview • Syllabus Quiz • Environment Setup (google colab/notebook, PyCharm, etc.)

Week 2: Jan 17 - Jan 23	• Introduction to Intelligent Systems • Student Group Creation for Final Project • Jan 19 MLK Day Holiday
Week 3: Jan 24 - Jan 30	• Role of AI/ML/Data Analytics in Intelligent Systems • Workshop on Decision-Making Process (Supervised ML) • Assignment 1
Week 4: Jan 31 – Feb 6	• Heuristic Programming: Greedy Best First Search • Breadth First Search (BFS) and Depth First Search (DFS) for Intelligent Systems
Week 5: Feb 7 - Feb 13	• Greedy Search and Hill Climbing • A* Tree Search for Intelligent Systems • Assignment 2
Week 6: Feb 14 - Feb 20	• Intro to ML for Intelligent Systems and SVM • Intelligent Systems Topic Presentation
Week 7: Feb 21 - Feb 27	• Exam 1 Syllabus Review and Exam Preparation: ◦ From Week 1 to Week 5 • Exam 1
Week 8: Feb 28 – Mar 6	• Decision Tree for Intelligent Systems • Regression Models for Intelligent Systems • Assignment 3
Week 9: Mar 7 - Mar 13	• Intro to Neural Networks for Intelligent Systems • Convolution Neural Network • CNN Workshop
Week 10: Mar 14 - Mar 20	• Mar 16- Mar 20: Spring Break
Week 11: Mar 21 - Mar 27	• Recurrent Neural Network • Final Group Project Proposal Presentation • Mar 25: Last day to withdraw
Week 12: Mar 28 – Apr 3	• Ethics and Ethical Issues of Intelligent Systems • Legal and Privacy Issues of Intelligent Technologies • Assignment 4

Week 13: Apr 4 - Apr 10	• Exam 2 Syllabus Review and Exam Preparation: ◦ Week 6 and From Week 8 to Week 11 • Exam 2
Week 14: Apr 11 – Apr 17	• Trustworthy AI for Multimodal Data • Case-study on Sensor and Raspberry Pi-based Intelligent Systems
Week 15: Apr 18 - Apr 24	• Intro to Intelligent Agents and Markov Chains • Markov Decision Process • Assignment 5
Week 16: Apr 25 – May 1	• Q-learning • Final Group Project Demo Presentation and Submission
Week 17: May 2 – May 8	• Exam 3 Syllabus Review and Exam Preparation: • Week 12 and From Week 14 to Week 16 • Exam 3

All content for this course is located in and can be accessed via Moodle, the Computer Science Content Management System, which can be found here: [CS3270: Intelligent Systems (Section 01) [S26] (https://courses.cs.westga.edu/course/view.php?id=1234)](https://courses.cs.westga.edu/course/view.php?id=1234).

Attendance Policy

All students are expected to attend and actively participate in every class and lecture. The instructor will closely monitor in-class engagement (e.g., discussions, conversations, Q&A sessions) and attendance. If a student cannot attend a class due to unavoidable circumstances, they must provide proper justification, such as a medical certificate or other valid documentation.

Missing in-class activities (e.g., sudden/surprise quizzes or coding exercises based on previous lectures) will affect the final grade. Throughout the semester, there will be five such sudden/unannounced class activities, each worth 1% of the final grade. Missing any of these will result in a 1% deduction from the total score for each missed activity (please see the Evaluation section for more details on class activities).

Course Policies and Resources

Attendance Policy:

- All students are expected to attend and actively participate in every class and lecture. The instructor will closely monitor in-class engagement (e.g., discussions, conversations, Q&A sessions) and attendance. If a student cannot attend a class due to unavoidable circumstances, they must provide proper justification, such as a medical certificate or other valid documentation.

- Missing in class activities (e.g., sudden/surprise quizzes or coding exercises based on previous lectures) will affect the final grade. Throughout the semester, there will be five such sudden/unannounced class activities, each worth 1% of the final grade. Missing any of these will result in a deduction of 1% from the total score for each missed activity (See the Evaluation section for more details on class activities).

Email and Announcement Checking:

- All-important announcements will be posted on Moodle throughout the semester. You will also receive these notifications via your UWG email. It is recommended that you regularly check both your email and Moodle for course updates.
- All schedules follow Eastern Time.
- Students are responsible for checking and keeping up with communications sent to their UWG account. **Email sent from another account,** other than your UWG account, will **not be answered.** Emails should be sent directly to the email account listed above. Emails sent via the class website on CourseDen may not be answered. Answers via email may take up to two business days. Please note that even though a student may have sent an email about an issue, you remain responsible for complying with all class policies and all homework deadlines without exception.

Assignment/Project/Exam/Quiz/Other Activities:

- This course values students' eagerness to learn over perfection. There will be no cheating allowed.
- No late submission will be allowed. However, exceptions can be considered based on the proposed justification.
- On deadline day and the day before the deadline, there will be no meeting or discussion regarding the assignments/projects/exams/quizzes, which will allow the students to learn how to plan for chasing a deadline. Therefore, if you have any questions, suggestions, or feedback on assignments/projects/exams/quizzes, please do it earlier the day before the deadline.
- There will not be any makeup for any assignments.
- There will be no drop lab exercises or project grades.
- All assignments are due at the scheduled time. Late work is not accepted.
- Each assignment can be submitted up to 5 times within the deadline.
- Exams' instructions will be provided via Moodle.
- It is the student's responsibility to make sure the correct file is submitted.
- Under no circumstances will submissions be accepted via email.
- **APIs and other supporting tools may require legacy support during certain activities. Students are responsible for determining which version or versions are needed to complete a specific task. For example, some APIs may not be supported in Python 3.9 but may work in Python 3.8; in such cases, students must identify the appropriate version.**

Academic Honesty/Integrity Statement

- Under no circumstances will the use of AI for report writing be allowed. Please be careful about plagiarism and cheating on your graded and non-graded activities and exercises.

- Do not share your code or assignments with others until the grading. However, you can discuss and do brainstorming with classmates without sharing code and actual assignment.
- Do not copy from classmates' work.
- In this course, Gen AI tools are permitted in all coursework. If there is any exception for a particular activity or assignment, that will be explicitly described in the assignment instructions. Please follow the instructions that will be described in each assignment.
- The students must understand the working principle of each line of code and be capable of justifying each line of code. Gen-AI can give a piece of code without knowing the entire requirements. Therefore, please be careful when using Gen-AI since it only kills time without the synergy of your assignment problem.
- If you use any sentences/text from any scientific research paper or publications, please properly cite them and use quote "".
- Students must obey the University of West Georgia honor code and maintain academic honesty.

IT Support or Internal Server Access:

- UWG and CAM school's policies will determine all kinds of UWG IT support or internal server access.
- If someone forgets the username or password or loses the credentials of any server/software/other facilities, it is advised to contact the support staff immediately.

Generative AI-based LLM Use Policy:

- Students may use Gen-AI to generate code for their projects since the IT/Software industries are encouraging this to save time. However, the students must understand the working principle of each line of code and be capable of justifying each line of code. Gen-AI can give a piece of code without knowing the entire requirements. Therefore, please be careful when using Gen-AI since it only kills time without the synergy of your particular assignment problem.

Special Needs:

Please see the Accessibility Services guidelines of the Institutional Policies section.

Generative Artificial Intelligence Course Policy

In this course, Gen AI tools are permitted in all coursework. If there is any exception for a particular activity or assignment, that will be explicitly described in the assignment instructions.

Students will evaluate the use of Gen AI critically and be aware of the limitations of Gen AI tools. Gen AI tools used to generate coursework must be clearly marked, cited, and acknowledged, as failure to do so could constitute plagiarism. Gen AI output must be supported, documented and cited with scholarly research from print sources and/or digital databases. The following links contain information on how to cite Generative AI material:

- APA Style Blog on citing ChatGPT
- MLA Style Center on citing Generative AI

- and per faculty guidelines

Any violations of these guidelines will be subject to the academic and disciplinary policies listed in the UWG Honor Code (see: Student Handbook)

College/School Policies

Dr. James 'Earl' Perry College of Mathematics, Computing, and Sciences (CMCS) offers transformative educational experiences that engage students in the latest research and technology.

Our students participate in relevant, real world research, projects, and internships in the lab, the field, and industry. They make interdisciplinary connections with mentors in community, corporations, government, and alumni partners to explore innovations in science, computing, and mathematics.

Students are encouraged to practice the following Big Six college experiences to be successful in CMCS coursework and degree programs:

(A) Connect with professors, staff, coaches, etc. who care about you as a person:

- Connect with a professor(s) who makes you excited to learn
- Connect with a mentor(s) who cares about you as a person
- Connect with a mentor(s) who pushes you to reach your goals

(B) Participate in experiential learning opportunities:

- Complete a long-term project such as a capstone project
- Participate in a high-impact practice such as undergraduate research or an internship
- Get involved in extracurricular activities and groups

Institutional Policies

Academic Support

UWG is committed to student success, and the following resources will help you be more successful in your classes.

Center for Academic Success: The Center for Academic Success (http://www.westga.edu/cas/) provides tutoring, academic coaching, and supplemental instruction to help all undergraduate students succeed academically. For more information, contact them: 678-839-6280 or cas@westga.edu.

University Writing Center: The University Writing Center (https://www.westga.edu/writing/) assists students with the writing process. For more information, contact them: 678-839-6513 or writing@westga.edu.

Accessibility Services: Students with a documented need/accommodation may work with UWG Accessibility and Testing Services to receive essential services specific to their needs. All accommodations are based on documentation and relevant statutes and regulations. If a student needs accommodations in a course, or if the student needs to make special arrangements in case the building must be evacuated,

the student should notify their instructor and provide a copy of their Student Accommodations Report (SAR), which is available only from Accessibility and Testing Services. Faculty cannot offer accommodations without receipt of the SAR, and retroactive accommodations are not permitted. For more information, please contact Accessibility and Testing Services (https://www.westga.edu/student-services/accessibility-testing/index.php).

Online Course Content

UWG takes students' privacy concerns seriously: technology-enhanced and partially and fully online courses use sites and entities beyond UWG and students have the right to know the privacy policies of these entities. For help with your online classes, additional online tutoring and other student success services, information on privacy and accessibility (https://uwgonline.service-now.com/kb?sys_kb_id=47b0ef89c3c58210ab7303bdc00131bc&id=kb_article_view&sysparm_rank=1&sysparm_tsqueryId=1ef80ce3c3e982507db3b2459901318a), and technology requirements, visit this UWG Online (https://uwgonline.service-now.com/kb/) Help site.

UWG's online virtual tutoring service is Tutor.com. Tutor.com provides 24/7, on-demand, 1-to-1 tutoring and homework help in more than 250 subjects. The expert tutors at Tutor.com can help students work through tough homework problems, improve their writing skills, study for a test, review difficult concepts, and so much more! Tutor.com can be accessed in CourseDen under the Resources dropdown menu and is available to all UWG students, regardless of course modality. More information can be found on UWG Online's Tutor.com: Tutoring Service Knowledge Base article (https://www.google.com/url?q=https://uwgonline.service-now.com/kb/?id%3Dkb_article_view%26sysparm_article%3DKB0010788&sa=D&source=docs&ust=1689091469862762&usg=AOvVaw2vhm-Y9CAGpzHoFZpHnqPF).

Students enrolled in online courses can find answers to many of their questions in the Online/Off-Campus Student Guide (http://uwgonline.westga.edu/online-student-guide.php).

Honor Code

At the University of West Georgia, we believe that academic and personal integrity are based upon honesty, trust, fairness, respect, and responsibility. Students at UWG assume responsibility for upholding the Honor Code. UWG students pledge to refrain from engaging in acts that do not maintain academic and personal integrity. These include, but are not limited to, plagiarism*, cheating*, fabrications*, aid of academic dishonesty, lying, bribery or threats, and stealing. When a student chooses to enroll at the University of West Georgia students pledge the following:

> *Having read the honor code of UWG, I understand and accept my responsibility to uphold the values and beliefs described, and to conduct myself in a manner that will reflect the values of the institution in such a way as to respect the rights of all UWG community members. As a UWG student, I will represent myself truthfully and complete all academic assignments honestly.*

I understand that if I violate this code, I will accept the penalties imposed, should I be found responsible for violations through the processes due to me as a University community member. These penalties may include expulsion from the University. I also recognize that my responsibility includes willingness to confront members of the University community, if I feel there has been a violation of the Honor Code.

For more information on the University of West Georgia Honor Code, please visit the Office of Community Standards (https://www.westga.edu/administration/vpsa/ocs/index.php) site.

UWG Email Policy

University of West Georgia students are provided a MyUWG e-mail account. The University considers this account to be an official means of communication between the University and the student. The purpose of the official use of the student e-mail account is to provide an effective means of communicating important university related information to UWG students in a timely manner. It is the student's responsibility to check their email.

Mental Health Support

If you or another student find that you are experiencing a mental health issue, free confidential services are available on campus in the Counseling Center. (https://www.westga.edu/student-services/counseling/) Students who have experienced sexual or domestic violence may receive confidential medical and advocacy services with the Patient Advocates in Health Services (https://www.westga.edu/student-services/health/). To report a concern anonymously, please go to UWGcares (https://www.westga.edu/uwgcares/).

Online counseling (https://www.westga.edu/student-services/counseling/index.php) is also available for online students.

ELL Resources

If you are a student having difficulty with English language skills, and / or U.S. culture is not your home culture, specialized resources are available to help you succeed. Please visit the E.L.L. resource page (https://www.westga.edu/isap/ell-resources.php) for more information.

Credit Hour Policy

The University of West Georgia grants one semester hour of credit for work equivalent to a minimum of one hour (50 minutes) of in-class or other direct faculty instruction AND two hours of student work outside of class per week for approximately fifteen weeks. For each course, the course syllabus will document the amount of in-class (or other direct faculty instruction) and out-of-class work required to earn the credit hour(s) assigned to the course. Out-of-class work will include all forms of credit-bearing activity, including but not limited to assignments, readings, observations, and musical practice. Where available, the university grants academic credit for students who verify via competency-based testing, that they have accomplished the learning outcomes associated with a course that would normally meet the requirements outlined above (e.g. AP credit, CLEP, and departmental exams).

Additional Items